QED3 assembly outline

Lattice positivity, renormalization, and an interacting Osterwalder–Schrader theory

Spatial volume tends to infinity at fixed lattice spacing; the continuum limit is taken afterwards.

LP Reflection positivity and Gaussian domination in lattice QED

For every real mass, cubic-volume limits have the sharp Gaussian bound, lattice symmetries, and reflection positivity in all nine mirror families.

MB Block averaging in Maxwell's theory

An exact Gaussian decomposition separates coarse fields and fluctuations, with local means, decaying covariances, and field-domain bounds. Iteration gives photon coordinates at every scale.

FB Block averaging with fermions

Exact Grassmann block integration with local bounds for propagators, reconstruction maps, determinants, and insertions.

FD The fermion determinant: quartic response

The free lattice quartic coefficient converges to its continuum value. A compact smooth probe has a strictly positive coefficient.

both

lattice laws and Gaussian bounds

sector trace argument

RP Consequences of extra reflection positivity

Reflection Hilbert spaces, positive translation semigroups, spectral-cone bounds, and analytic continuation. The OS axioms yield relativistic reconstruction.

LF The renormalization group: large fields

Uniform tail bounds at every scale for the normalized zero-flux interacting measure. Given a uniformly bounded holomorphic continuation in the complex sources, analytic interpolation and Cauchy's formula control derivatives of the large-field remainder.

RS The renormalization group: one step

Integrating one fluctuation scale gives an effective density of the same form. Extraction and rescaling contract the inherited remainder; new terms and their differences satisfy uniform bounds.

free coefficient and positive probe

general RP consequences

R Rotation invariance from reflection positivity

Translation invariance, coordinate and diagonal reflection positivity, and a controlled exponential growth bound imply rotation covariance of Schwinger functions away from coincidences.

C Clustering from reflection positivity

Local correlations approach their invariant part at a rate bounded by region size divided by separation. In ergodic components the limit is the product of expectations.

large-field control

RT The renormalization group: full trajectory

One microscopic mass and vacuum assignment gives controlled trajectories on the reference tori. Iteration yields the exact terminal representation of the small-field integral.

terminal identity and tail lemma

CL The continuum limit and the Osterwalder–Schrader axioms

Volume-first continuum subsequential limits exist, with convergence of all smeared correlation functions and joint cumulants. Complex-source generating functions and their logarithms near zero converge locally uniformly; Cauchy's formula passes their derivatives to the limit. Almost every ergodic component retains closedness and the sharp Gaussian bound and satisfies the OS axioms, with covariance away from coincidences.

UV Ultraviolet stability of the partition function

A common bulk subtraction gives bounds uniform in lattice spacing. The full defect between the two reference partition functions is bounded, including photon and flux contributions.

OS axioms and component bounds

IR Infrared stability of the partition function

Two-sided pressure bounds hold uniformly in outer volume. Reflection and folding give normalized transfer bounds for products of local observables.

transfer

FP The interacting theory: quartic response

Complex sources and Cauchy estimates control the reference response through fourth degree in each cell. Products of Wick squares, and separately Wick fourth powers, satisfy cutoff-uniform bounds on the prescribed separated grid at arbitrary outer volume, including all flux sectors.

finite trace and reflection estimates

all-volume product estimates

AS An interacting Osterwalder–Schrader theory

For sufficiently small positive charge e, almost every ergodic component of every prescribed volume-first limit satisfies the OS axioms and

$$\left|\kappa_4^{\mu\epsilon}(f) - e^4 b(f)\right| \le C_f e^{4+\sigma}, \qquad \kappa_4^{\mu\epsilon}(f) \ge \tfrac{1}{2} e^4 b(f) > 0.$$

Here f is the compact probe from FD, $b(f) > 0$, and $\sigma > 0$. Both conclusions hold on one full-measure component set for each fixed parent, charge, and probe.

Construction manuscript or final assembly External general theorem

Arrows show principal dependencies. Joined inputs are required together; crossings without a dot do not join.

The program and its endpoint

	Code	Manuscript or result
1	**MB**	Block averaging in Maxwell's theory
2	**FB**	Block averaging with fermions
3	**RS**	The renormalization group: one step
4	**RT**	The renormalization group: full trajectory
5	**LF**	The renormalization group: large fields
6	**UV**	Ultraviolet stability of the partition function
7	**IR**	Infrared stability of the partition function
8	**FD**	The fermion determinant: quartic response
9	**FP**	The interacting theory: quartic response
	LP	Reflection positivity and Gaussian domination in lattice QED
	CL	The continuum limit and the Osterwalder–Schrader axioms
External	**RP**	Consequences of extra reflection positivity
External	**R**	Rotation invariance from reflection positivity
External	**C**	Clustering from reflection positivity
Assembly	**AS**	An interacting Osterwalder–Schrader theory

Order of limits. At each fixed lattice spacing $a > 0$, first let the spatial size tend to infinity and choose an infinite-volume limit. Then take a subsequential limit as $a \downarrow 0$.

Final result. Fix an admitted positive reference mass and chamber parameters. For sufficiently small positive physical charge e, the common microscopic parameters selected by **RT** give volume-first continuum subsequential limits of the all-flux curvature laws, with convergence of all smeared correlation functions and joint cumulants. For every such parent, almost every translation-ergodic component μ_ξ is a closed real tempered field, retains the sharp Gaussian bound, and has Schwinger functions satisfying the OS axioms, including clustering.

For the fixed compact smooth probe f supplied by **FD**, the same component set satisfies

$$\left|\kappa_4^{\mu_\xi}(f) - e^4 b(f)\right| \le C_f e^{4+\sigma}, \qquad \kappa_4^{\mu_\xi}(f) \ge \tfrac{1}{2} e^4 b(f) > 0.$$

Here κ_4 is the connected fourth moment of the smeared curvature, $b(f) > 0$, and $\sigma > 0$. The components are non-Gaussian. Both conclusions hold on one full-measure component set for each fixed parent, charge, and probe.

Source material: QEM1, field-strength theorem; VTM1, component cumulant theorem.

Block averaging and renormalization

MB Block averaging in Maxwell's theory

Inputs. The Maxwell quadratic form, gauge fixing, and block geometry.

Main theorem. An exact normalized Gaussian decomposition separates the photon field into a coarse background and independent fluctuations. The constrained means, covariances, and local inverses satisfy uniform locality, decay, regularity, and field-domain bounds. Iteration gives photon fluctuation coordinates at every scale.

Source material: MBM1; photon coordinates in TPM1.

FB Block averaging with fermions

Inputs. **MB** and the chamber fermion operator in an admissible complex background connection.

Main theorem. An exact Grassmann block-integration identity gives uniform local bounds for propagators, fluctuation covariances, and reconstruction maps. Determinant contributions and marked insertions satisfy the bounds used in scale iteration and terminal integration.

Source material: FBM1; reconstruction in RNM1; terminal estimates in TPM1.

RS The renormalization group: one step

Inputs. **MB**, **FB**, an admitted running mass, and a small local interaction on the specified field domain.

Main theorem. Integrating one restricted fluctuation scale produces an exact effective density of the same form, with updated mass, vacuum energy, background scalar terms, and interaction. Extraction and rescaling contract the normalized inherited remainder. Newly generated terms and their differences satisfy the iteration bounds. The local transformation agrees on admissible covers.

Source material: RNM1, NLM1, and the local comparison estimates in CVM1.

RT The renormalization group: full trajectory

Inputs. **RS**.

Main theorem. For each cutoff, one microscopic mass and vacuum-energy assignment gives a controlled trajectory through the stopping scales on every torus in the reference family. The assignment is independent of the external probe and harmonic twist. Iteration expresses the small-field integral exactly as a terminal measure, retaining sources, bounded measurable observables, and normalization factors.

Source material: CVM1; terminal measure identity in TPM1.

Large fields and partition stability

LF The renormalization group: large fields

Inputs. Photon coordinates from **MB** and the finite-volume trace construction from **LP**.

Main theorem. The normalized zero-flux interacting measure assigns small probability to the excluded fluctuation and terminal-curvature configurations on the reference volumes, uniformly in the ultraviolet cutoff and the initial real mass and vacuum parameters. Given a holomorphic continuation of the excluded contribution with a uniform complex-source bound, its smallness for real sources extends by analytic interpolation to a smaller complex neighborhood. Cauchy's formula then controls its source derivatives.

Source material: TPM1, zero-flux domination and terminal tails; analytic interpolation and Cauchy estimates in RRM1.

UV Ultraviolet stability of the partition function

Inputs. **RT**, **LF**, and terminal fermion estimates from **FB**.

Main theorem. After a common cutoff-dependent bulk subtraction, the reference partition functions satisfy bounds uniform in lattice spacing. At fixed physical charge, the full defect between the two reference partition functions is bounded uniformly in the ultraviolet cutoff, including the photon determinant and magnetic-flux contributions.

Source material: Partition estimates in TPM1; full partition defect in VTM1, §3.

IR Infrared stability of the partition function

Inputs. The reference partition bound from **UV** and the trace, reflection, and folding identities from **LP**.

Main theorem. One common bulk subtraction gives upper and lower bounds on the logarithm of the partition function per physical volume, uniformly over arbitrarily large admissible tori. Folding gives a normalized transfer inequality for products of local observables, with normalization cost controlled by the reference partition defect.

Source material: VTM1, §2 and the general transfer proposition.

FD The fermion determinant: quartic response

Inputs. The free chamber symbol, heavy-mode gap, Schur complement, and inverse estimates from **FB**.

Main theorem. The fourth logarithmic derivative of the free lattice fermion determinant converges to its continuum value. The second and fourth derivatives of the normalized determinant ratio satisfy uniform bounds. There is a real compact smooth curvature probe f with strictly positive continuum quartic coefficient $b(f)$.

Source material: PRM1, positive-probe and free-response theorems.

The four-point function and continuum limits

FP The interacting theory: quartic response

Inputs. **RT**, **FB**, **LF**, **FD**, and **IR**.

Main theorem. The source-shift identity expresses Wick correlations as derivatives of a normalized complex-source generating function. On the reference torus, analytic estimates and Cauchy's formula bound the difference between its mixed derivatives and their free values, through degree four in each cell. The resulting bounds for finite products of Wick squares, and separately Wick fourth powers, transfer to distinct cells of the prescribed separated grid on arbitrarily large admissible tori. The bounds are uniform in the cutoffs and include all magnetic-flux sectors. The source identity and terminal logarithm estimates supply the complex bound used in **LF** to differentiate the large-field remainder.

Source material: Complex-source identities and response estimates in RRM1; all-volume Wick-product bounds in VTM1.

LP Reflection positivity and Gaussian domination in lattice QED

Inputs. The all-flux chamber lattice model at fixed lattice parameters.

Main theorem. For every real mass, the cubic finite-volume curvature laws are tight. Every subsequential limit along integer cubic side lengths tending to infinity obeys the sharp Gaussian bound, lattice translations and cubic symmetries, and reflection positivity in the three coordinate and six diagonal mirror families. The reflection form is positive for all square-integrable observables supported strictly on either side.

Source material: LPM1, main theorem.

CL The continuum limit and the Osterwalder–Schrader axioms

Inputs. **LP**, **RP**, **R**, and **C**.

Main theorem. Volume-first continuum subsequential limits exist, and *all smeared correlation functions and joint cumulants converge* along the chosen limiting sequences. For each finite family of Schwartz tests, the complex-source generating functions converge locally uniformly, as do their logarithms on a common zero-free neighborhood of zero. Gaussian domination supplies the uniform complex bounds; Cauchy's formula passes source derivatives to the limit.

Each prescribed parent is a closed real tempered two-form law with the sharp Gaussian bound. Almost every translation-ergodic component retains the same Gaussian constant and the reflection forms. Its Schwinger functions satisfy OS regularity, Euclidean covariance away from coincidences, reflection positivity, permutation symmetry, and clustering.

Source material: Complex-source convergence and continuum existence in CEM1; component inheritance in ECM1; supported tensor tests in QEM1.

External theorems and assembly

RP Consequences of extra reflection positivity

Inputs. Translation invariance, coordinate and diagonal reflection forms, and the field-insertion bound $\mathbb{E}e^{F(f)} \le \exp\{C(\|f\|_1 + \|f\|_2^2)\}$.

Main theorem. Coordinate reflection positivity gives reflection Hilbert spaces and positive translation semigroups. Diagonal reflection positivity gives spectral-cone bounds and identifies the zero-energy projections with the invariant part. The exponential bound gives analytic translations, continuation of correlation kernels, and extension to polynomial and supported tensor tests. The OS axioms yield relativistic reconstruction.

Source material: Reflection and analytic arguments in RPM1; tensor-test and reconstruction arguments in QEM1 and QCM1.

R Rotation invariance from reflection positivity

Inputs. **RP**, translation invariance, coordinate and diagonal reflection positivity, the exponential bound in **RP**, and an orthogonal tensor representation with planar weights $|w| \le 2$.

Main theorem. The Schwinger functions are covariant under continuous rotations away from coincident configurations.

Source material: RPM1, Theorem A.

C Clustering from reflection positivity

Inputs. Translation invariance, invariance and reflection positivity in the nine coordinate and diagonal orientations, and the spectral results of **RP**.

Main theorem. Local correlations approach their translation-invariant conditional part. For observables supported in a cube of radius r and separated by $|a|_\infty \ge 4r$, the error is bounded by a universal constant times $r/|a|_\infty$ and the product of their L^2 norms. Qualitative mixing holds for all L^2 observables. For an ergodic law the limit is the product of expectations.

Source material: The clustering appendix in QCM1; invariant projections in ECM1.

AS An interacting Osterwalder–Schrader theory

Inputs. **CL** and **FP**, with the parameters from **RT** and the probe from **FD**.

Main theorem. For sufficiently small positive charge, almost every ergodic component of every prescribed volume-first parent satisfies the OS axioms and the positive fourth-cumulant estimate on page 2. Both conclusions hold on one full-measure component set for each fixed parent, charge, and probe.

Source material: Component cumulants in VTM1; assembly in QEM1.

Where the existing material goes

Existing material	Destination and role
MBM1	**MB**: exact photon decomposition and local estimates.
FBM1	**FB**: fermion integration, inverses, determinants, and background estimates.
RNM1	**RS**: extraction and contraction; general reconstruction bounds move with **FB**.
NLM1	**RS**: exact nonlinear fluctuation integration and its remainder estimates.
CVM1	**RT**: common tuning and controlled trajectories; local cover compatibility accompanies **RS**.
TPM1	**MB**: iterated photon coordinates. **FB**: general terminal fermion estimates. **RT**: exact terminal identity. **LF**: zero-flux domination and excluded-field tails. **UV**: reference partition comparison.
RRM1	**FP**: complex-source identities, Cauchy estimates, and the interacting remainder. **LF**: analytic interpolation of tail bounds.
PRM1	**FD**: free second/fourth response and the positive probe.
VTM1	**UV**: full photon/flux partition defect. **IR**: pressure bounds and general observable transfer. **FP**: all-flux Wick products and conditional component transfer. **AS**: application to the selected continuum components.
LPM1	**LP**: lattice construction and reflection positivity.
CEM1	**CL**: compactness, prescribed-parent recovery, and convergence of correlations and cumulants by complex sources and Cauchy's formula.
ECM1	**CL**: component reflection forms and sharp Gaussian bound. **RP**, **C**: general cone, invariant-projection, and clustering results.
RPM1	**R**: rotations. **RP**: spectral and analytic results.
QCM1	**C**: quantitative clustering. **RP**: analytic continuation and reconstruction.
QEM1	**CL**: supported tensor tests. **AS**: parameter admission and assembly.